Introduction

1.1 Background

The rapid evolution of digital technologies has transformed the way individuals, organizations, and governments communicate, store information, and conduct daily operations. While these technological advancements have improved productivity and connectivity, they have also significantly increased the number and sophistication of cyber threats. Cybercriminals exploit vulnerabilities in computer systems and networks through attacks such as unauthorized access, phishing, insider threats, ransomware, malware infections, and data breaches. These incidents often result in financial losses, operational disruptions, reputational damage, and compromise of sensitive information.

Digital Forensics is a specialized branch of cybersecurity that focuses on the identification, collection, preservation, examination, analysis, and presentation of digital evidence. It enables investigators to reconstruct cyber incidents, identify attackers, understand attack patterns, and provide legally admissible evidence for judicial proceedings. Traditionally, digital forensic investigations involve manual examination of system logs, network traffic, user activities, and security events. However, with the exponential growth of data generated by modern information systems, manual investigation has become increasingly time-consuming, resource-intensive, and susceptible to human error.

Machine Learning (ML) has emerged as a promising solution for automating various stages of digital forensic investigations. By learning patterns from historical data, ML algorithms can identify anomalies, classify malicious activities, prioritize security incidents, and assist forensic analysts in making informed decisions. Unlike conventional rule-based systems that depend on predefined attack signatures, ML-based systems are capable of recognizing previously unseen or evolving attack behaviors through statistical learning techniques.

This project proposes the development of an AI-Powered Digital Forensics and Cyber Incident Analysis Platform using Machine Learning. The proposed platform focuses on analyzing cybersecurity logs from the CERT Insider Threat Dataset, one of the most widely used benchmark datasets for insider threat research. The system aims to automate log analysis, detect suspicious user activities, classify cyber incidents using classical machine learning algorithms, and generate meaningful forensic insights through interactive visualization and reporting.

Unlike many modern cybersecurity solutions that rely on computationally expensive deep learning models, this project exclusively employs classical machine learning techniques such as Isolation Forest, Random Forest, Decision Tree, and Logistic Regression. These algorithms provide interpretable results, require lower computational resources, and are well suited for academic research and practical deployment in environments with limited hardware capabilities.

The proposed platform seeks to improve the efficiency of forensic investigations by minimizing manual analysis, reducing incident response time, and assisting cybersecurity professionals in identifying potential insider threats with greater accuracy. Through the integration of data preprocessing, anomaly detection, machine learning, visualization, and automated reporting, the system demonstrates the practical application of artificial intelligence in modern digital forensic investigations.

1.2 Problem Statement

Organizations generate massive volumes of security logs every day from authentication systems, network devices, web servers, file systems, and endpoint monitoring tools. These logs contain valuable evidence regarding user behavior, system activities, and potential security incidents. However, manually analyzing such large-scale datasets to identify malicious activities is an extremely challenging task.

Traditional forensic investigation techniques primarily depend on expert analysts who manually inspect log files to identify suspicious events. This approach suffers from several limitations, including excessive investigation time, difficulty in identifying hidden attack patterns, increased false positives, and limited scalability for large organizations. Furthermore, insider threats often exhibit subtle behavioral changes that may remain undetected using conventional signature-based detection methods.

Although several commercial Security Information and Event Management (SIEM) solutions provide automated log management, they are often expensive, complex to configure, and unsuitable for educational or small-scale environments. Therefore, there is a need for an intelligent, lightweight, and cost-effective forensic analysis platform capable of automatically detecting suspicious activities using machine learning techniques while maintaining transparency and explainability.

The proposed project addresses these challenges by developing an AI-assisted digital forensic platform capable of analyzing user activity logs from the CERT Insider Threat Dataset, identifying anomalous behaviors, classifying potential cyber incidents, and supporting forensic investigators with actionable insights.

1.3 Objectives

The primary objective of this project is to develop an intelligent digital forensic platform that assists cybersecurity professionals in identifying suspicious activities through machine learning techniques.The specific objectives are:

- To study the concepts of digital forensics, cyber incident analysis, and insider threat detection.
- To analyze the CERT Insider Threat Dataset and understand its characteristics.
- To preprocess and clean the dataset for machine learning applications.
- To perform feature engineering and data transformation for improved model performance.
- To implement classical machine learning algorithms for anomaly detection and incident classification.
- To compare the effectiveness of different machine learning algorithms in detecting suspicious activities.
- To develop an interactive dashboard for visualizing forensic evidence and security incidents.
- To generate automated forensic reports that assist investigators during incident response.
- To build a scalable platform that can be extended for real-world digital forensic investigations.

1.4 Scope of the Project

The scope of the proposed project is limited to the analysis of insider threat activities using the CERT Insider Threat Dataset. The platform focuses on identifying suspicious user behavior by analyzing log files generated from user authentication, web browsing, file access, removable device usage, and email activities.

The project includes data preprocessing, exploratory data analysis, feature engineering, machine learning model training, anomaly detection, attack classification, dashboard visualization, and automated forensic report generation.

The project does not include live network packet capture, malware reverse engineering, memory forensics, or deep learning-based detection techniques. Instead, it emphasizes explainable and computationally efficient machine learning algorithms suitable for educational research and practical implementation.

1.5 Proposed Solution

The proposed solution is an AI-powered forensic investigation platform that integrates data analysis, machine learning, and visualization into a unified workflow. The platform begins by collecting cybersecurity logs from the CERT Insider Threat Dataset. The collected data undergoes preprocessing steps such as missing value handling, duplicate removal, feature encoding, timestamp conversion, and normalization.

After preprocessing, meaningful features are extracted and supplied to machine learning models for anomaly detection and attack classification. Algorithms such as Isolation Forest are employed to identify abnormal user behaviors, while Random Forest, Decision Tree, and Logistic Regression are used to classify suspicious activities based on learned patterns.

The prediction results are further analyzed to calculate risk levels and generate interactive dashboards displaying incident timelines, activity distributions, user behavior statistics, and anomaly trends. Finally, the platform produces structured forensic reports summarizing the detected incidents, associated evidence, prediction outcomes, and recommended actions for investigators.

This integrated approach enables faster investigation, improved detection accuracy, and reduced manual effort compared to traditional forensic analysis methods.

1.6 Expected Outcomes

Upon successful completion of the project, the proposed platform is expected to provide the following outcomes:

- Detection of anomalous user activities using machine learning techniques.
- Classification of cyber incidents with improved accuracy.
- Interactive visualization of security events through dashboards.
- Automated generation of forensic investigation reports.
- Improved efficiency in digital forensic investigations by reducing manual analysis.
- A lightweight, explainable, and scalable forensic analysis platform suitable for academic research and future enhancements.

Supporting literature

2.1 Literature Review

Paper 1: Threats from Unintentional Insiders: An Assessment

Modern organizations increasingly face security risks originating not only from malicious attackers but also from employees whose unintentional actions compromise organizational security. This study examines the causes, characteristics, and impacts of unintentional insider threats. The authors identify that careless handling of confidential information, weak password practices, accidental disclosure of sensitive data, and failure to follow cybersecurity policies contribute significantly to security incidents.

The paper emphasizes that traditional security mechanisms mainly focus on external attacks and often overlook behavioural anomalies exhibited by legitimate users. Consequently, organizations require intelligent monitoring systems capable of analysing user activities and identifying abnormal behavioural patterns before they escalate into serious incidents.

Although the paper provides valuable insights into insider threat behaviour, it primarily focuses on identifying the causes of insider threats rather than proposing an automated detection mechanism. This limitation motivates the development of intelligent machine learning-based forensic systems capable of continuously analysing user activities and detecting suspicious behaviour.

Key Findings

- Insider threats are difficult to identify using traditional security tools.
- User behaviour analysis plays a critical role in insider threat detection.
- Continuous monitoring improves organizational security.
- Automated analysis is required for handling large-scale organizational data.

Limitation

The study does not implement or evaluate a machine learning model for automated insider threat detection.

Paper 2: Fronesis: Digital Forensics-Based Early Detection of Ongoing Cyber-Attacks

The Fronesis framework proposes an intelligent digital forensic system capable of identifying cyber attacks during their early stages. The framework integrates digital forensic evidence collection with security event analysis to improve incident response time and forensic readiness.

The study demonstrates how security logs generated from multiple system components can be correlated to reconstruct attack scenarios. Automated analysis enables investigators to detect suspicious activities more efficiently compared to traditional manual investigations.

The proposed framework significantly reduces investigation time while improving the accuracy of forensic analysis. However, the framework mainly focuses on forensic event correlation and does not extensively evaluate different machine learning algorithms for anomaly detection.

Key Findings

- Digital forensic readiness reduces incident response time.
- Automated evidence correlation improves investigation quality.
- Continuous monitoring enhances cyber defence. Early detection minimizes attack impact.

Limitation

Limited comparison of multiple machine learning algorithms for detecting insider threats.

Paper 3: Insider Threat Detection Model Using Anomaly-Based Isolation Forest Algorithm

This research proposes an anomaly detection model based on the Isolation Forest algorithm for identifying insider threats. Unlike supervised learning methods, Isolation Forest isolates anomalous observations by randomly partitioning data, making it highly suitable for detecting unknown attack patterns.

The model was evaluated using insider threat datasets and demonstrated promising performance in identifying abnormal user behaviour with relatively low computational complexity. The study highlights the effectiveness of anomaly detection for cybersecurity applications where labelled attack data may be limited.

The findings strongly support the use of Isolation Forest in digital forensic investigations, particularly for identifying suspicious user activities within organizational environments.

Key Findings

- Isolation Forest effectively detects anomalous user behaviour.
- Suitable for high-dimensional cybersecurity datasets.
- Efficient for large-scale log analysis.
- Requires fewer computational resources than many complex models.

Limitation

The study focuses only on anomaly detection and does not perform detailed attack classification or forensic report generation.

Paper 4: Performance Evaluation Framework for Insider Threat Detection Using Machine Learning

This study presents a comprehensive framework for evaluating the performance of various machine learning algorithms in detecting insider threats within organizational environments. The primary objective of the framework is to establish a standardized methodology for comparing machine learning models based on detection accuracy, computational efficiency, precision, recall, F1-score, and false positive rate. The authors emphasize that selecting an appropriate machine learning algorithm is essential for developing reliable insider threat detection systems capable of handling large volumes of organizational log data.

The framework utilizes publicly available insider threat datasets to train and evaluate multiple machine learning classifiers. Various preprocessing techniques, including data cleaning, feature engineering, and normalization, are applied before model training to improve prediction performance. The study compares several classical machine learning algorithms and concludes that ensemble-based methods generally achieve better overall performance due to their robustness and ability to reduce overfitting.

One of the significant contributions of this research is the introduction of a systematic evaluation methodology rather than focusing solely on a single algorithm. By comparing multiple performance metrics, the framework enables researchers and practitioners to select models that best suit their specific cybersecurity applications. The study also highlights that explainable machine learning techniques are particularly valuable in digital forensic investigations, where analysts require transparent reasoning behind prediction outcomes.

The findings of this research directly support the methodology proposed in the present project. Since the proposed AI-Powered Digital Forensics and Cyber Incident Analysis Platform employs multiple machine learning algorithms—including Isolation Forest for anomaly detection and Random Forest, Decision Tree, and Logistic Regression for classification—the evaluation framework described in this paper provides an appropriate foundation for comparing model performance and selecting the most effective algorithm for forensic analysis.

Key Findings

- Provides a standardized framework for evaluating insider threat detection models.
- Demonstrates the importance of comparing multiple performance metrics rather than relying solely on accuracy.
- Highlights the effectiveness of ensemble learning methods such as Random Forest.
- Emphasizes proper preprocessing and feature engineering for improving prediction performance.
- Supports explainable machine learning for digital forensic investigations.

Limitation

The proposed framework focuses primarily on algorithm evaluation and performance comparison. It does not integrate visualization dashboards, automated forensic reporting, or a complete end-to-end digital forensic investigation platform.

2.2 Findings and Proposal

The literature review indicates that existing research has made significant progress in applying artificial intelligence and machine learning to cybersecurity and digital forensics. Behaviour-based analysis, anomaly detection, and automated evidence correlation have emerged as effective approaches for identifying cyber incidents and insider threats. In particular, Isolation Forest has proven suitable for detecting previously unseen malicious activities, while Random Forest and Decision Tree classifiers have demonstrated strong performance in attack classification tasks.

Despite these advancements, several research gaps remain. Many existing studies focus on a single aspect of cyber incident detection, such as anomaly detection or behavioural analysis, without integrating the entire digital forensic investigation process. Several frameworks are

conceptual or evaluated using limited datasets, and few provide investigators with an interactive platform that combines data preprocessing, anomaly detection, attack classification, visualization, and forensic report generation. Furthermore, many recent solutions rely on deep learning models, which often require significant computational resources and provide limited interpretability, making them less suitable for academic environments and forensic investigations where explainability is essential.

To address these limitations, the proposed project introduces an AI-Powered Digital Forensics and Cyber Incident Analysis Platform using Machine Learning. The platform utilizes the CERT Insider Threat Dataset as its primary data source and employs classical machine learning algorithms—including Isolation Forest for anomaly detection and Random Forest, Decision Tree, and Logistic Regression for incident classification. The system integrates data preprocessing, feature engineering, exploratory data analysis, machine learning prediction, dashboard visualization, and automated forensic report generation into a single workflow.

Unlike existing approaches that emphasize only detection accuracy, the proposed platform aims to support the complete forensic investigation process by providing investigators with actionable insights, explainable predictions, and structured reports. By relying on computationally efficient and interpretable machine learning techniques, the proposed system offers a practical, scalable, and academically suitable solution for digital forensic investigations.

2.3 Literature Review Summary

Paper	**Focus Area**	**ML Technique**	**Dataset**	**Strength**	**Research Gap**
Threats from Unintentional Insiders	Insider Threat Analysis	Behaviour Analysis, NARX	CERT R4.2	Identifies behavioural indicators of insider threats.	No automated forensic investigation platform.
Fronesis	Digital Forensics	Event Correlation	System and Network Logs	Early cyber-attack detection through forensic readiness.	Limited ML-based insider threat detection.
Insider Threat Detection Model	Anomaly Detection	Isolation Forest	Insider Threat Dataset	Efficiently detects unknown behavioural anomalies.	No attack classification or forensic reporting.
Performance Evaluation Framework	Threat Detection	Random Forest, Decision Tree, Logistic Regression, SVM	Public Insider Threat Dataset	Provides comparative evaluation of multiple ML models using standard performance metrics.	No visualization or integrated forensic investigation platform.

Table 2.1: Summary of Literature Review

System Analysis

Machine Learning-based digital forensic systems require a systematic analysis of the available datasets before applying machine learning algorithms. The quality of the dataset directly influences the performance of anomaly detection and classification models. Therefore, it is essential to understand the characteristics of the dataset, identify its important features, and perform exploratory data analysis before preprocessing and model development. This chapter presents the analysis of the CERT Insider Threat Dataset, followed by data preprocessing techniques, feature analysis, data visualization, machine learning algorithms, project pipeline, and feasibility analysis used in the proposed AI-Powered Digital Forensics and Cyber Incident Analysis Platform.

3.1 Analysis of the Dataset

The CERT Insider Threat Dataset is one of the most widely used benchmark datasets for insider threat detection research. It was developed by the CERT Division of Carnegie Mellon University to simulate realistic organizational environments and user activities. The dataset contains multiple log files representing different aspects of employee behaviour, including user authentication, web browsing, removable device usage, email communication, and file access. These logs collectively provide detailed information that enables researchers to analyse user behaviour, detect suspicious activities, and evaluate the performance of machine learning models in cybersecurity applications.

The complete CERT Insider Threat Dataset consists of six primary log files, each representing a different category of user activity within an organization. Since the objective of this initial project report is to perform exploratory data analysis rather than develop the complete system, only three representative datasets—logon.csv, device.csv, and email.csv—were selected for detailed analysis. These datasets were chosen because they represent three of the most important behavioural indicators used in insider threat detection: user authentication activities, removable device usage, and email communication. The remaining datasets, namely http.csv, file.csv, and LDAP, will be incorporated during the implementation phase of the proposed forensic investigation platform.

Table 3.1: CERT Insider Threat Dataset Files Used in the Project

Dataset File	**Description**	**Used**
`logon.csv`	Records user login and logout activities.	Yes
`device.csv`	Records USB device connection and removal activities.	Yes
`email.csv`	Contains employee email communication records.	Yes
`http.csv`	Stores employee web browsing activities.	No
`file.csv`	Records file creation, modification, and deletion activities.	No
`LDAP.csv`	Contains employee organizational information.	No

3.2 Explore the Dataset

Before applying any machine learning techniques, an exploratory analysis was conducted to understand the characteristics of the selected datasets. Exploratory Data Analysis (EDA) provides valuable insights into the structure, quality, and distribution of data. It also helps identify missing values, duplicate records, feature types, and overall dataset composition, enabling appropriate preprocessing before model training.

The exploratory analysis was performed using Google Colab with the Python programming language and the Pandas library. Three representative datasets—logon.csv, device.csv, and email.csv—were loaded and examined individually.

Dataset Dimensions

The first step in the exploratory analysis involved identifying the number of records and attributes available in each selected dataset. This information provides an understanding of the size of the dataset and the amount of information available for analysis.

```
Logon Dataset:  (854859, 5)
Device Dataset: (495380, 5)
Email Dataset:  (68631, 11)
```

Figure 3.1: Dataset Dimensions of Selected CERT Datasets

Dataset Preview

The first five records of the logon.csv dataset were examined to understand the structure of the dataset and the information stored in each feature. This preview provides an overview of user login activities recorded within the organizational environment.

1 to 5 of 5 entries Filter

index	id	date	user	pc	activity
0	{X1D9-S0ES98JV-5357PWMI}	01/02/2010 06:49:00	NGF0157	PC-6056	Logon
1	{G2B3-L6EJ61GT-2222RKSO}	01/02/2010 06:50:00	LRR0148	PC-4275	Logon
2	{U6Q3-U0WE70UA-3770UREL}	01/02/2010 06:53:04	LRR0148	PC-4124	Logon
3	{I0N5-R7NA26TG-6263KNGM}	01/02/2010 07:00:00	IRM0931	PC-7188	Logon
4	{D1S0-N6FH62BT-5398KANK}	01/02/2010 07:00:00	MOH0273	PC-6699	Logon

Figure 3.2: First Five Records of logon.csv

Dataset Information

The dataset information was analysed to determine the total number of records, feature names, data types, and non-null values. This information is useful for identifying data quality issues and planning appropriate preprocessing techniques.

```
logon.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 854859 entries, 0 to 854858
Data columns (total 5 columns):
 #   Column    Non-Null Count   Dtype
---  ------    --------------   -----
 0   id        854859 non-null  object
 1   date      854859 non-null  object
 2   user      854859 non-null  object
 3   pc        854859 non-null  object
 4   activity  854859 non-null  object
dtypes: object(5)
memory usage: 32.6+ MB
```

Figure 3.3: Dataset Information of logon.csv

Summary of Selected CERT Datasets

Dataset	Records	Features	Purpose
logon.csv	854,859	5	User Login Behaviour
device.csv	495,380	5	USB Device Monitoring
email.csv	68,631	11	Email Communication

Table 3.2: Datasets Used for Analysis

3.3 Data Preprocessing

Data preprocessing is an essential stage in any machine learning project because the quality of input data directly affects the performance of the developed models. Raw datasets often contain inconsistencies such as missing values, duplicate records, incorrect data types, and irrelevant attributes that can reduce prediction accuracy. Therefore, preprocessing techniques were applied to prepare the selected datasets for further analysis and machine learning implementation.

For this project, the preprocessing stage was performed using Python and the Pandas library in Google Colab. The selected datasets—logon.csv, device.csv, and email.csv—were examined and processed through missing value analysis, duplicate record analysis, data type verification, date format conversion, and feature selection.

Missing Value Analysis

Missing values can negatively affect machine learning algorithms by introducing incomplete information and reducing prediction reliability. Therefore, each selected dataset was examined to identify null or missing records.

The analysis revealed that logon.csv, device.csv, and email.csv did not contain missing values. As a result, no data imputation or record removal was required during preprocessing.

Dataset	Missing Values
logon.csv	0
device.csv	0
email.csv	0

Table 3.3: Missing Values in the Selected Datasets

Duplicate Record Analysis

Duplicate records increase data redundancy and may bias machine learning models. To ensure data quality, duplicate records were identified and removed from the selected datasets.

The analysis showed that all three datasets contained zero duplicate records. Although no duplicate entries were found, duplicate checking was performed as a standard preprocessing step to ensure dataset integrity.

Dataset	Duplicate Records
logon.csv	0
device.csv	0
email.csv	0

Table 3.4: Duplicate Records in the Selected Datasets

Feature Selection

Feature selection was performed to identify the most relevant attributes required for analysing user behaviour and detecting insider threats. Selecting only meaningful features reduces computational complexity while improving the efficiency of machine learning models.

Dataset	Selected Features	Purpose
logon.csv	user, pc, activity, date	Analyse user authentication behaviour.
device.csv	user, pc, activity, date	Monitor removable device activities.
email.csv	user, from, to, date	Analyse employee email communication.

Table 3.5: Selected Features from the Datasets

The selected features represent the core behavioural information required for the proposed AI-powered digital forensic platform. These attributes will be used during the implementation phase for anomaly detection, behavioural analysis, and cyber incident classification.

Data type Verification

Before applying machine learning algorithms, the data types of all features were verified. This process helps identify categorical, textual, and temporal attributes that require further preprocessing.

The analysis indicated that most attributes, such as user, pc, activity, and email-related fields, were stored as object (string) data types, while the date attribute represented timestamp information. Verifying the data types ensured that appropriate transformations could be applied before model training.

```
print(logon.dtypes)

id                       object
date          datetime64[ns]
user                     object
pc                       object
activity                 object
dtype: object
```

Figure 3.4: Data Type Verification of logon.csv

Data Visualisation

Data visualization is an important stage in digital forensic analysis because it transforms raw log data into meaningful graphical representations. Visualizing user activities helps investigators quickly identify behavioural patterns, abnormal activities, and trends that may indicate potential insider threats. In this project, bar charts were generated using Python's Matplotlib library in Google Colab to analyse authentication events, computer usage, and removable device activities. The visualizations provide an overview of employee behaviour and support the interpretation of the selected CERT Insider Threat datasets before machine learning model

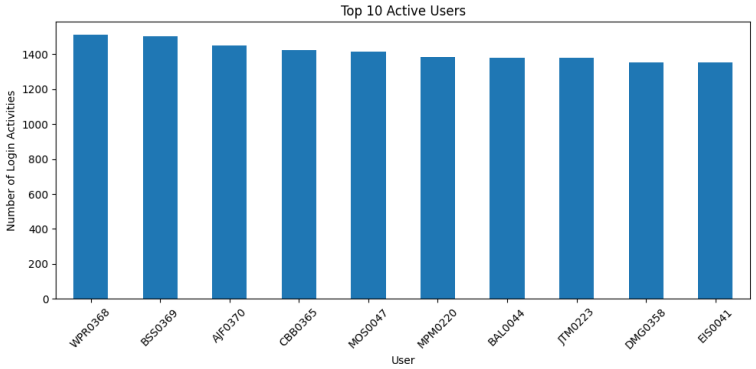


Figure 3.5: Top 10 Active Users

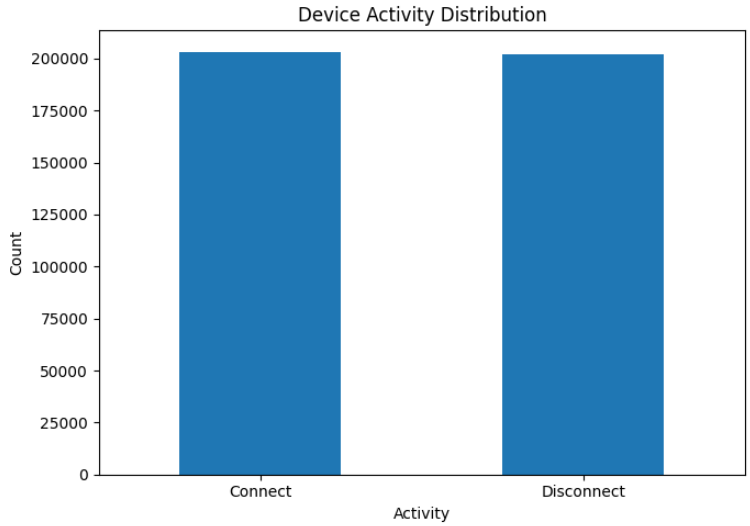


Figure 3.6: Top 10 Computers by User Activity

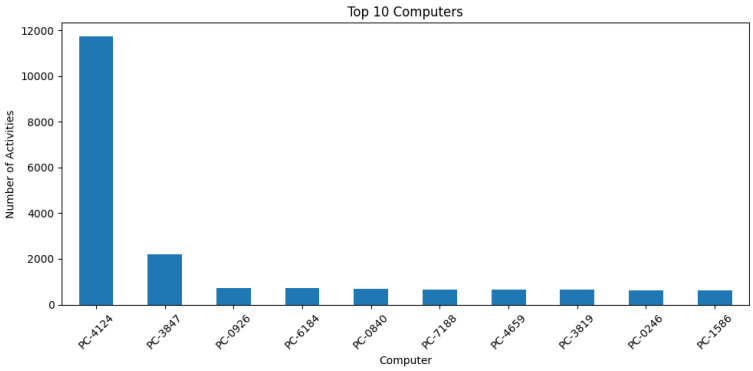


Figure 3.7: Device Activity Distribution

3.4 Proposed System Architecture

The proposed AI-Powered Digital Forensics Investigation Support Platform is designed to automate the analysis of digital forensic evidence using machine learning techniques. The system integrates data collection, preprocessing, feature extraction, anomaly detection, classification, visualization, and forensic report generation into a unified workflow. The objective is to assist cybersecurity analysts in identifying suspicious user activities and improving the efficiency of digital forensic investigations.

The system begins by loading user activity logs from the CERT Insider Threat Dataset. The selected datasets—logon.csv, device.csv, and email.csv—are first preprocessed to ensure consistency and data quality. During preprocessing, missing values and duplicate records are verified, timestamp attributes are standardized, and relevant features are selected for analysis.

After preprocessing, the prepared data undergoes feature engineering, where important behavioural attributes are extracted for machine learning analysis. The processed data is then supplied to the selected machine learning algorithms. The Isolation Forest algorithm is employed to identify anomalous user behaviour, while Random Forest, Decision Tree, and Logistic Regression are utilized to classify suspicious activities and estimate potential insider threats.

The prediction results are presented through an interactive visualization dashboard, al-

lowing investigators to examine user behaviour, system activities, and detected anomalies. Finally, the platform generates a structured forensic investigation report summarizing the findings and supporting decision-making during cyber incident investigations.

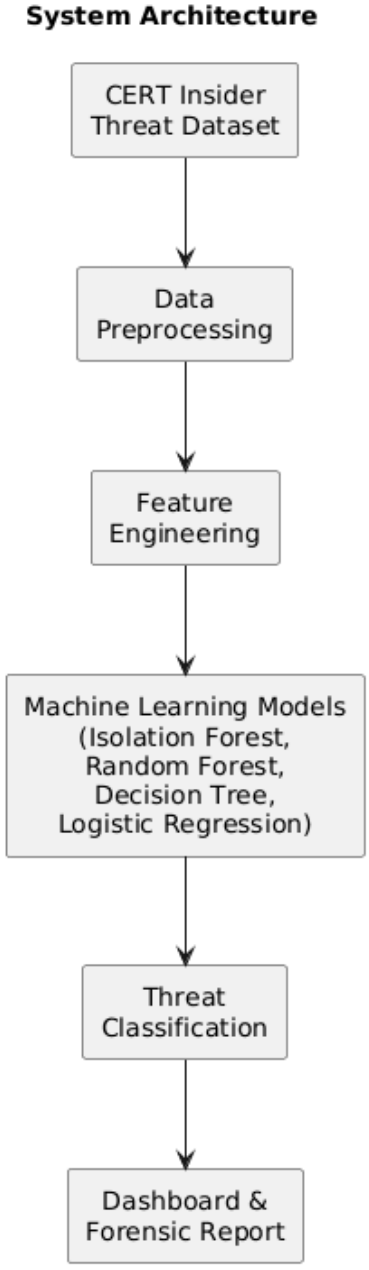


Figure 3.8: Device Activity Distribution

3.5 Machine Learning Algorithm

Machine learning algorithms play an important role in analysing behavioural patterns within digital forensic datasets. The proposed platform employs a combination of anomaly detection and supervised classification algorithms to improve the identification of suspicious activities while maintaining model interpretability and computational efficiency.

Isolation Forest

Isolation Forest is an unsupervised anomaly detection algorithm specifically designed to identify unusual observations within large datasets. Unlike traditional classification algorithms, Isolation Forest isolates anomalous records by randomly selecting features and partitioning the dataset into decision trees. Since anomalies differ significantly from normal observations, they require fewer partitions to become isolated.

In the proposed platform, Isolation Forest is used to identify abnormal user behaviour from authentication logs, device activities, and communication records. The algorithm is particularly suitable because the CERT Insider Threat Dataset initially contains no predefined anomaly labels.

Advantages

- Efficient for large datasets.
- Detects previously unseen anomalies.
- Does not require labelled training data.
- Computationally efficient for insider threat detection.

Random Forest

Random Forest is an ensemble learning algorithm that constructs multiple decision trees and combines their predictions to improve classification accuracy. The algorithm reduces overfitting while providing reliable predictions for complex datasets.

Within the proposed platform, Random Forest will classify suspicious user activities after behavioural features have been extracted. The algorithm's robustness and high predictive performance make it suitable for cybersecurity applications.

Advantages

- High classification accuracy.
- Resistant to overfitting.
- Handles high-dimensional datasets.
- Robust against noisy data.

Desicion Tree

Decision Tree is a supervised learning algorithm that classifies data by creating hierarchical decision rules based on feature values. The algorithm is highly interpretable because each prediction can be explained through a sequence of decision nodes.

In the proposed system, Decision Tree assists investigators by providing explainable classifications that support digital forensic investigations and security audits.

Advantages

- Easy to understand.
- Provides explainable decisions.
- Fast training and prediction.
- Suitable for categorical features

Logistic Regression

Logistic Regression is a statistical classification algorithm commonly used for binary prediction problems. The algorithm estimates the probability that a particular observation belongs to a specific class.

Within the proposed forensic platform, Logistic Regression serves as a baseline classification model for distinguishing normal and suspicious behavioural activities.

Advantages

- Simple implementation.
- Fast execution.
- Easy interpretation.
- Effective for binary classification.

3.6 Target Variable Analysis

The selected CERT Insider Threat datasets used in this project do not contain predefined target labels indicating whether an activity is normal or suspicious. Since the proposed system employs machine learning for insider threat detection, a target variable will be generated during the implementation phase. The generated labels will represent behavioural classes such as Normal Activity and Suspicious Activity, based on anomaly scores produced by the Isolation Forest algorithm and further classification using supervised learning models. This approach enables the development of predictive models while preserving the original integrity of the CERT dataset.

Target Label	**Description**
Normal	Represents legitimate user behaviour within the organization.
Suspicious	Represents anomalous activities that may indicate an insider threat.

Table 3.6: Target Labels Used for Classification

3.7 Project Pipeline

The proposed AI-powered digital forensic investigation platform follows a structured processing pipeline to analyse organizational user activities. The pipeline begins with the collection of digital forensic logs from the CERT Insider Threat Dataset. The selected datasets are then preprocessed to remove inconsistencies and prepare the information for machine learning analysis.

Following preprocessing, feature engineering extracts meaningful behavioural attributes from authentication logs, device activities, and email communications. These features are supplied to multiple machine learning algorithms, where Isolation Forest performs anomaly detection while Random Forest, Decision Tree, and Logistic Regression classify suspicious user behaviour.

The prediction results are then visualized through an interactive dashboard that presents user activity summaries, anomaly trends, and threat classifications. Finally, the platform automatically generates a forensic investigation report containing detected anomalies, behavioural summaries, and recommended actions for cybersecurity investigators.

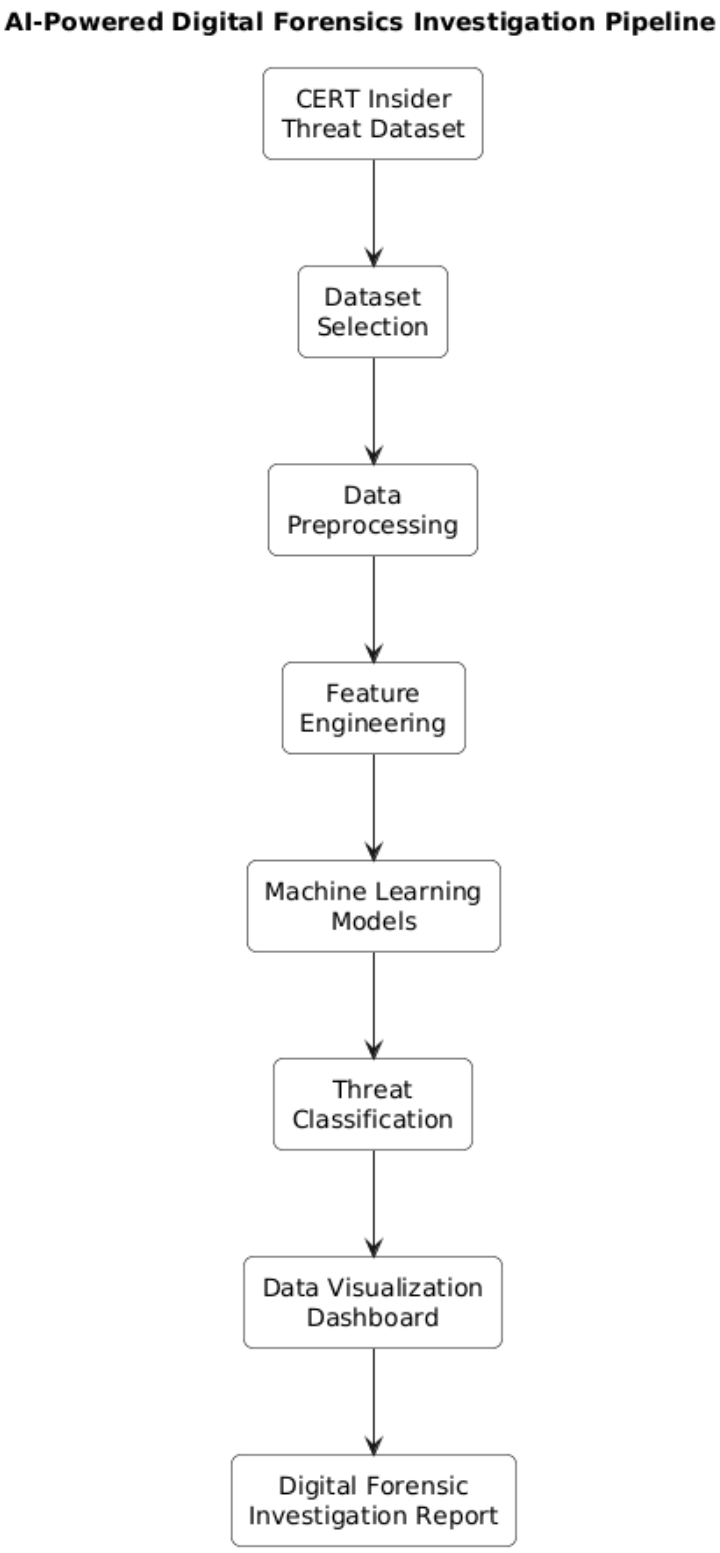


Figure 3.9: Project Pipelines